The Inverse-GCD Operator Q_N: Definitions, a Restricted Rayleigh Bound, and Withdrawal of Full-Spectrum Floor Claims

Q6 in proper form — spectral note, not a prize paper

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Disclaimer. This note defines the inverse-GCD ("Q6") matrices and proves one restricted inequality (Bridge*). It does *not* claim Goldbach, the Riemann Hypothesis, or Navier–Stokes regularity. Full-spectrum floors $\lambda_{\min} > -1/2$ are *false* for the named matrices.

Status

Claim	Status
Definitions of Q_N, $\widetilde{Q}_N$, H_N	**Locked**
Bridge*: $R(e_p - e_q) > -1/2$ on $\widetilde{Q}_N$	**Proved**
Positive orthant: $v \geq 0 \Rightarrow v^\top \widetilde{Q}_N v \geq 0$	**Proved**
$\lambda_{\min}(Q_N) > -1/2$ for all N	**False**
$\lambda_{\min}(\widetilde{Q}_N) > -1/2$ for all N	**False**
Dark-state $\Leftrightarrow$ Goldbach (June 2026 draft)	**Withdrawn**
Multi-representation Bridge*	**Open**

Abstract

We give a corrected account of the inverse-GCD operators used in earlier "Q6" drafts. Write

$$Q_N(i,j) = \frac{1}{\gcd(i,j)}, \qquad \widetilde{Q}_N(i,j) = \frac{1}{\gcd(i,j)\sqrt{ij}}, \qquad 1 \leq i,j \leq N,$$

and let H_N be the degree-normalized form of $\widetilde{Q}_N$. We prove that for distinct primes p, q,

$$R(e_p - e_q) = \frac{1}{2}\Big(\frac{1}{p^2} + \frac{1}{q^2}\Big) - \frac{1}{\sqrt{pq}} \; > \; -\frac{1}{2}$$

on $\widetilde{Q}_N$ (Bridge*). We also record that every nonnegative vector has nonnegative Rayleigh quotient on $\widetilde{Q}_N$. We withdraw the full-spectrum Bridge claim $\lambda_{\min} > -1/2$: already for moderate N, $\lambda_{\min}(Q_N) < -1/2$ and $\lambda_{\min}(\widetilde{Q}_N) < -1/2$. Earlier dark-state / Goldbach packaging and any Navier–Stokes claim from these matrices are withdrawn.

1 Operators

Definition 1.1 (Raw inverse-GCD)**.**

$$Q_N(i,j) = \frac{1}{\gcd(i,j)}, \qquad 1 \le i,j \le N.$$

Definition 1.2 (Normalized inverse-GCD)**.**

$$\widetilde{Q}_N(i,j) = \frac{1}{\gcd(i,j)\sqrt{ij}}.$$

In particular, for distinct primes $p, q \le N$,

$$\widetilde{Q}_N(p,p) = \frac{1}{p^2}, \qquad \widetilde{Q}_N(p,q) = \frac{1}{\sqrt{pq}}.$$

Definition 1.3 (Degree-normalized form)**.** *Let $d_i = \sum_{j=1}^{N} \widetilde{Q}_N(i,j)$ and $D = \operatorname{diag}(d_1,\dots,d_N)$. Set*

$$H_N = D^{-1/2}\,\widetilde{Q}_N\,D^{-1/2}.$$

Remark 1.4 (Hygiene)**.** *These three matrices are different. A spectral statement about one does not automatically transfer to another, nor to any fluid commutator called "Theorem H" in companion notes.*

Remark 1.5 (Withdrawn identity)**.** *Some June 2026 drafts asserted $1/n = \sum_{d|n} \mu(d)\varphi(d)/d^2$. This fails already at $n = 2$ ($1/2$ versus $3/4$). No argument in this note uses that identity.*

2 What fails: full-spectrum floors

Proposition 2.1 (Full-spectrum Bridge is false)**.** *There exist N such that*

$$\lambda_{\min}(Q_N) < -1/2 \qquad \text{and} \qquad \lambda_{\min}(\widetilde{Q}_N) < -1/2.$$

In particular the universal claims $\lambda_{\min}(Q_N) > -1/2$ and $\lambda_{\min}(\widetilde{Q}_N) > -1/2$ for all N are false.

Numeric certificate. Direct eigendecomposition yields, for example, $\lambda_{\min}(Q_{10}) \approx -1.90 < -1/2$ and $\lambda_{\min}(\widetilde{Q}_{20}) \approx -0.505 < -1/2$. Reproducible by the accompanying script `scripts/bridge_floor_verify.py`. □

Remark 2.2. *A claimed universal lower bound $\lambda_{\min}(H_N) \ge -3/14$ is likewise not uniform in small N; see the companion verification suite. We do not package any H_N floor as a theorem here.*

3 Bridge*: single prime pair

Theorem 3.1 (Bridge*, single pair)**.** *Let $p \ne q$ be primes and $N \ge \max(p,q)$. For $v = e_p - e_q$,*

$$R(v) := \frac{v^\top \widetilde{Q}_N v}{\|v\|_2^2} = \frac{1}{2}\Big(\frac{1}{p^2} + \frac{1}{q^2}\Big) - \frac{1}{\sqrt{pq}} \;>\; -\frac{1}{2}.$$

Proof. Since $\|v\|_2^2 = 2$,

$$v^\top \widetilde{Q}_N v = \widetilde{Q}_N(p,p) + \widetilde{Q}_N(q,q) - 2\widetilde{Q}_N(p,q) = \frac{1}{p^2} + \frac{1}{q^2} - \frac{2}{\sqrt{pq}},$$

hence

$$R = \frac{1}{2}\Big(\frac{1}{p^2} + \frac{1}{q^2}\Big) - \frac{1}{\sqrt{pq}}.$$

Then

$$R + \frac{1}{2} = \frac{1}{2} + \frac{1}{2p^2} + \frac{1}{2q^2} - \frac{1}{\sqrt{pq}} > \frac{1}{2} - \frac{1}{\sqrt{pq}}.$$

Distinct primes force $pq \geq 6$, so $1/\sqrt{pq} \leq 1/\sqrt{6} < 1/2$. Therefore $R + 1/2 > 0$. $\square$

Remark 3.2. *The quotient may be negative: $(p,q) = (2,3)$ gives $R \approx -0.228 > -1/2$. The bound is a* restricted *floor on difference vectors of single prime pairs, not a full-spectrum eigenvalue floor.*

4 Positive orthant

Proposition 4.1 (Nonnegative cone)**.** *If $v \in \mathbb{R}^N$ satisfies $v_i \geq 0$ for all i, then $v^\top \widetilde{Q}_N v \geq 0$.*

Proof. Every entry of $\widetilde{Q}_N$ is positive, so the quadratic form on the nonnegative orthant is a sum of nonnegative terms. $\square$

5 What is withdrawn

Earlier public drafts used "Q6" as a route to Strong Goldbach via a dark-state lemma and a full-spectrum Bridge floor, and bundled the same operator into Navier–Stokes / RH packaging. Those claims are withdrawn.

1. **Broken detector.** The indicator difference $v_k(j) = 1_{\mathbb{P}}(j) - 1_{\mathbb{P}}(k-j)$ vanishes on genuine Goldbach pairs $p + (k-p) = k$, so it does not detect them.
2. **False sign lemma.** On raw Q_N, for $v = e_p - e_q$, $\langle v, Q_N v\rangle = 1/p + 1/q - 2$ is typically negative (e.g. $(3,7) \mapsto -1.52$). A claimed equivalence "dark state $\Leftrightarrow$ no Goldbach" does not hold as written.
3. **No NS map.** A spectral floor on Q_N, $\widetilde{Q}_N$, or H_N is not, by itself, a bound on $\|(u\cdot\nabla)u\|$ for Leray–Hopf solutions on $\mathbb{T}^3$.

The archive DOIs of those drafts may remain as dated history; they are not the claim of this note. The corrected public credit for this operator is Theorem 3.1 and Propositions 2.1–4.1.

6 Open problem

Open Problem (Multi-representation Bridge*)**.** *For even $k \leq N$ let $\mathcal{R}(k) = \{p \text{ prime} : 2 \leq p \leq k/2,\ k-p \text{ prime}\}$ and $v_k = \sum_{p\in\mathcal{R}(k)}(e_p - e_{k-p})$ (indices $\leq N$). Prove or disprove the existence of an absolute $c_0 < 1/2$ such that*

$$R(v_k) = \frac{v_k^\top \widetilde{Q}_N v_k}{\|v_k\|_2^2} \geq -c_0$$

for all such k, whenever $v_k \neq 0$.

Remark 6.1. *Numeric checks through* $N = 200$ *remain above* $-1/2$ *on sampled multi-rep vectors; that is evidence, not a proof.*

Reproducibility

```
python3 scripts/bridge_floor_verify.py 50
python3 -m unittest tests.test_bridge_star_h_n -v
```

Acknowledgments

This note supersedes the Goldbach-packaged Q6 preprint (10.5281/zenodo.20405589) as the author's intended public claim for the inverse-GCD operator.

References

[1] J.R. Simons, *A Q6 Spectral Route to the Strong Goldbach Conjecture* (withdrawn as a closure claim; archive only), Zenodo 10.5281/zenodo.20405589.

[2] J.R. Simons, *Bridge** *proof draft*, repository note `docs/BRIDGE-STAR-PROOF.md` (2026).